

Report: Network Setup and Connectivity Testing

Objective

The objective of this practical was to set up a local Wi-Fi network, test communication between devices using the **ping** command, analyze the network path using the **tracert** command, and identify connected devices using the **arp -a** command.

Step 1: Network Setup

1. Connected the Windows laptop to the home Wi-Fi network.
2. Connected an Android phone to the same Wi-Fi network.
3. Verified that both devices received IP addresses automatically from the router.

Device IP Addresses

Device	IP Address
Wi-Fi Router (Default Gateway)	192.168.1.1
My Laptop	192.168.1.100
Android Phone	192.168.1.101
Friend's Phone	192.168.1.102
Smart TV	192.168.1.103
Wireless Printer	192.168.1.104

Step 2: Testing Connectivity Using Ping

Opened **Command Prompt** on the laptop and executed:

```
ping 192.168.1.101
```

Sample Output

Pinging 192.168.1.101 with 32 bytes of data:

Reply from 192.168.1.101: bytes=32 time=3ms TTL=64

Reply from 192.168.1.101: bytes=32 time=2ms TTL=64

Reply from 192.168.1.101: bytes=32 time=3ms TTL=64

Reply from 192.168.1.101: bytes=32 time=2ms TTL=64

Ping statistics:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Observation: The laptop successfully communicated with the Android phone, confirming that both devices were connected to the same network.

Step 3: Testing the Route Using Traceroute

Executed the following command:

```
tracert www.youtube.com
```

Sample Output

Tracing route to www.youtube.com [142.250.183.206]

1	<1 ms	<1 ms	<1 ms	192.168.1.1
2	5 ms	5 ms	6 ms	ISP Gateway
3	9 ms	8 ms	9 ms	ISP Core Router
4	15 ms	14 ms	15 ms	Google Network
5	20 ms	21 ms	20 ms	142.250.183.206

Trace complete.

Number of Hops: 5

Observation: The command displayed the path taken by data packets from the laptop to the YouTube server through different routers.

Step 4: Disconnecting the Phone and Testing Again

Turned off Wi-Fi on the Android phone and executed:

```
ping 192.168.1.101
```

Sample Output

Pinging 192.168.1.101 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Observation: Since the phone was disconnected from Wi-Fi, the laptop could not communicate with it.

Step 5: Viewing Connected Devices

Executed:

```
arp -a
```

Sample ARP Table

Internet Address	Physical Address	Type
192.168.1.1	84-16-f9-12-34-56	Dynamic

Internet Address	Physical Address	Type
192.168.1.100	10-ab-cd-11-22-33	Dynamic
192.168.1.101	34-ef-56-78-90-ab	Dynamic
192.168.1.102	56-12-34-ab-cd-ef	Dynamic
192.168.1.103	98-76-54-32-10-aa	Dynamic
192.168.1.104	12-34-56-78-9a-bc	Dynamic

Observation: The ARP table displayed the IP addresses and corresponding MAC addresses of devices that had communicated on the local network.

Result

- Successfully connected all devices to the same Wi-Fi network.
- Verified connectivity using the **ping** command.
- Identified the network path using the **tracert** command.
- Observed **0% packet loss** when the phone was connected.
- Observed **100% packet loss** after disconnecting the phone.
- Displayed the connected devices and their IP/MAC addresses using the **arp -a** command.

Conclusion

The practical successfully demonstrated the setup of a local area network and the use of networking commands such as **ping**, **tracert**, and **arp -a**. The ping command verified communication between devices, traceroute showed the path to the destination server, and the ARP table identified devices on the network. These tools are essential for troubleshooting and maintaining computer networks.